

WESTERN SYDNEY UNIVERSITY



Project Report

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*Assignment 2 for COMP7023 Predictive Analytics
School of Computer, Data and Mathematical Sciences
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1 Introduction

Global warming has been the hot topic in the World Health Organisation (WHO) for many years. Vehicle emissions are a major source of environmental pollution and climate change, with carbon dioxide (CO₂) and nitrogen oxides (NO_x) being among the most significant pollutants. According to European Environment Agency (2023), CO₂ has impacted the increase of temperature, while NO_x emissions are associated with smog formation, acid rain, and adverse respiratory effects. Hence, the Intergovernmental Panel on Climate Change (IPCC) has suggested to closely monitor the gas emissions from the road vehicles.

This report analyses emissions using the data from the UK Vehicle Certification Agency (VCA), which provides verified measurements of vehicle fuel consumption and emissions under the Worldwide Harmonised Light Vehicle Test Procedure (WLTP). The VCA dataset ensures consistency and accuracy across vehicle testing in accordance with European and international standards (UK Vehicle Certification Agency 2024). Through statistical and predictive modelling, this study explores the key factors influencing emission levels to support sustainable transportation research and evidence-based environmental policymaking.

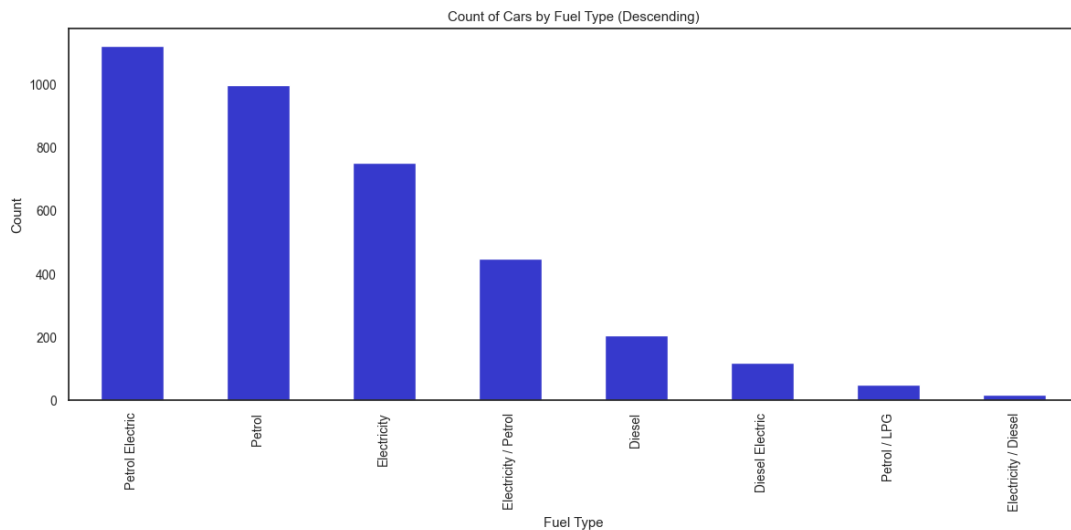
2 Exploratory Data Analysis

The data set contains of 3702 vehicles from different companies and 44 columns of measurement of vehicle types, fuel consumption and emission.

Category	Columns
Vehicle	Manufacturer, Model, Description, Transmission, Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS), Engine Power (Kw)
Fuel Consumption	Diesel VED Supplement, Testing Scheme, WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined, WLTP Imperial Combined (Weighted), WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined, WLTP Metric Combined (Weighted)
Electric Consumption	Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard	Euro Standard
Emmission	WLTP CO ₂ , WLTP CO ₂ Weighted, Emissions CO [mg/km], THC Emissions [mg/km], Emissions NO _x [mg/km], THC + NO _x Emissions [mg/km], Particulates [No.] [mg/km], RDE NO _x Urban, RDE NO _x Combined, Noise Level dB(A)
Date	Date of change

Category	Columns
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Out of the 44 columns only 9 columns which are Manufacturer, Model, Description, Transmission, Manual or Automatic, Fuel Type, Powertrain, Euro Standard, and Testing Scheme are the non-numeric data type, the rest of columns are numeric or boolean which can be used modelling. From following chart, we can see the distribution of fuel type of the vehicle in the data set. We can group the vehicles into three main categories: Fuel, Hybrid and Electric.

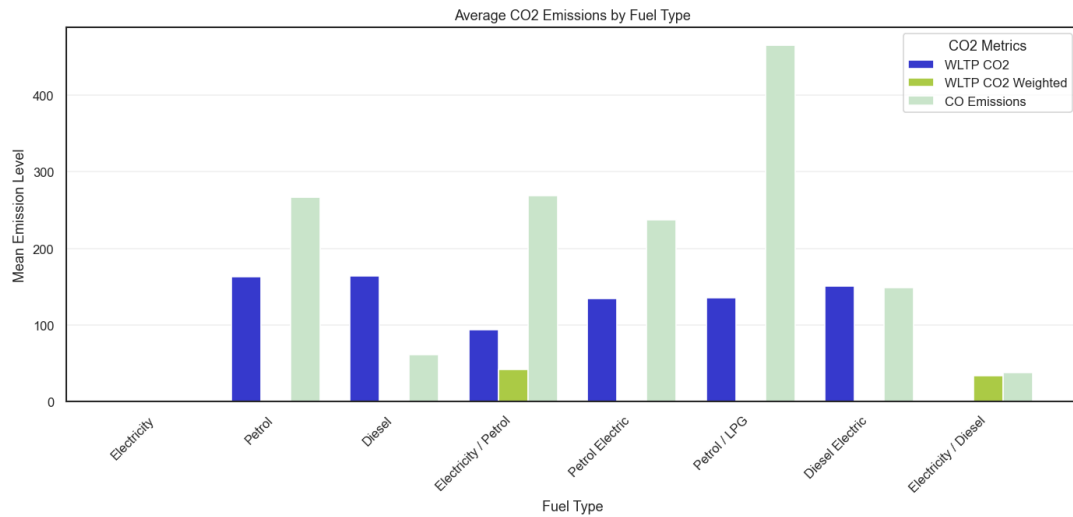


2.1 Emissions Analysis

As the discussion is about emission, we will further understand the different types of emissions such as Carbon Monoxide (CO), Carbon Dioxide (CO₂), Nitrogen Oxides (NO_x) and Total Hydrocarbons (THC). These emissions are not polluting the environment but impacting our health. According to WHO (2021), when CO binds with haemoglobin more readily than oxygen, it might reduce oxygen transport in blood and posing serious health risks to humans. While CO₂ is non-toxic to humans but it significantly contributes to global warming and climate change as it can trap the heat in the atmosphere (Intergovernmental Panel on Climate Change [IPCC], 2021). When the temperature increases, it might bring disasters to humans such as flood or extreme heat wave.

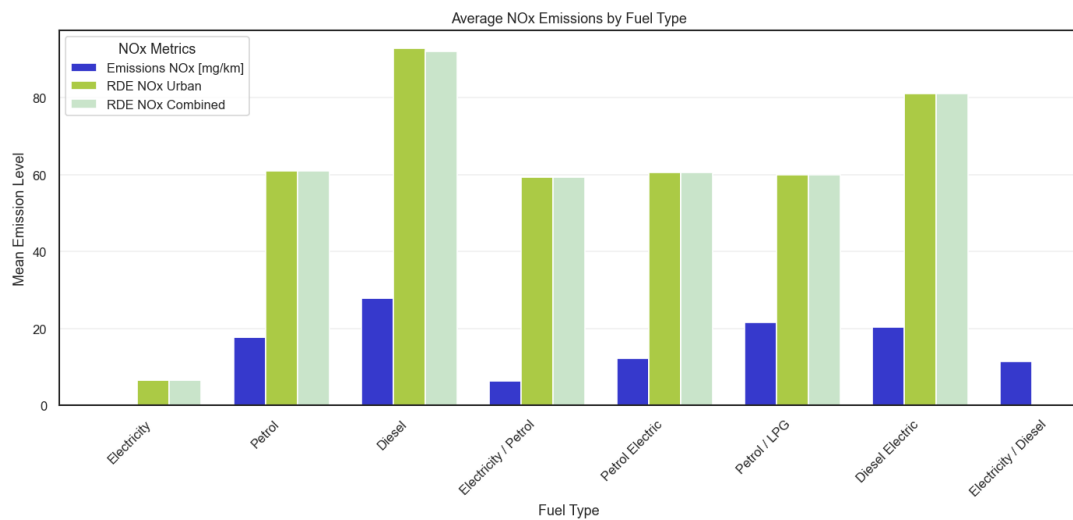
2.1.1 Carbon Monoxide (CO) and Carbon Dioxide (CO₂)

From the following bar chart, we can clearly see the emissions of CO₂ and CO co-exist in petrol or hybrid vehicles but not in electric cars. Emission of CO is high especially in diesel vehicles and emission of CO₂ based on WLTP CO₂ is about the same in petrol or diesel vehicles but low in hybrid vehicles. WLTP CO₂ Weighted seems to be not reliable as it is empty for most of the vehicles type.



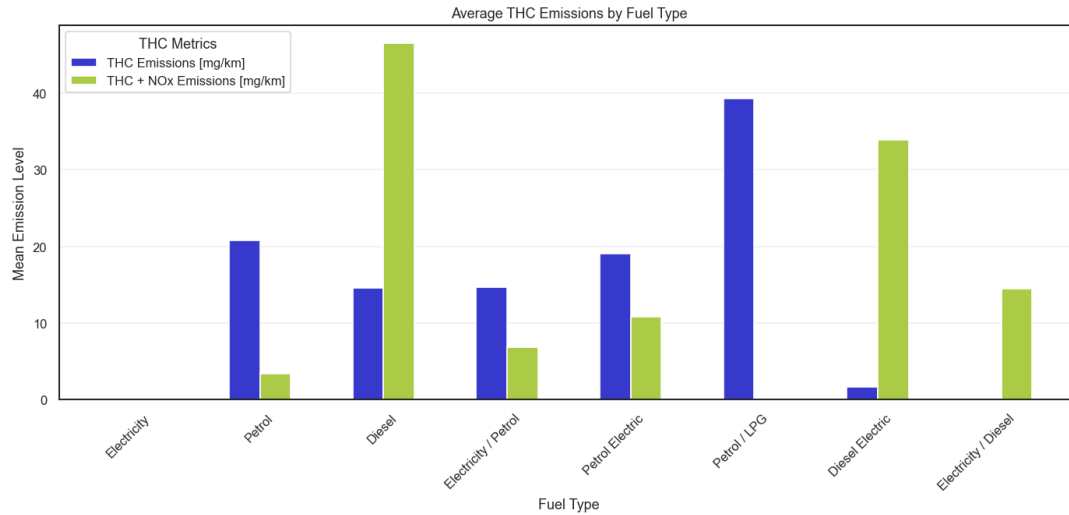
2.1.2 Nitrogen Oxides (NOx)

Nitrogen Oxides (NOx) emissions are a major cause to photochemical smog, acid rain, and respiratory health issues (U.S. Environmental Protection Agency [EPA], 2023). Based on following chart, we find NOx emission exists in all the vehicles including electric car. Electric cars do produce a small average amount of NOx according Veza et al (2023). However, it should be understood that the NOx production from electric vehicles is caused from the source of electric used to charge the car batteries. NO emissions from electric vehicles originate from the fossil fuel-based electricity used to charge them. The mean for NOx in Real Driving Emissions (RDE) for urban and combined are about same but they are higher than Emissions (NOx) [mg/km] which was the result tested in lab.



2.1.3 Total Hydrocarbons (THC)

Total Hydrocarbons (THC) are volatile organic compounds (VOCs) that react with NO_x in the presence of sunlight to form ground-level ozone and smog (European Environment Agency [EEA], 2022). THC might cause photochemical pollution and impact the urban air quality but it have to react with NO_x, hence, in this report we will not focus in analysing the relationship of THC with other features.



2.2 Target Variables

From the analysis, we have enough data for modelling across different fuel types but we will choose CO₂ as global warming is most common topic when discussing about the impacts of emission. As mentioned above, WLTP CO₂ Weighted is lacking of data distribution across fuel types, hence it will be excluded from target variables. Besides of CO₂, we will pick Emissions NO_x [mg/km] as our target variables because it has difference mean across fuel type which is easier for our comparison and prediction afterward in this report.

Based on the graphs above, electric cars have no CO₂ emissions, and low NO_x emissions. Therefore, all of the electric cars will be excluded from the datasets when we build the model. Given the distribution, two datasets will be created based on their fuel types - fuel cars only (data_fuel), and hybrid cars only (data_hybrid). Furthermore, since the target variables (WLTP CO₂ and NO_x emissions) are not highly included in Electricity / Diesel Fuel type, they will be removed from the data_hybrid.

3 Data Preprocessing

3.1 Model Building

After identified the target variables - Emissions NO_x [mg/km] and WLTP CO₂, based on the category that we have defined above, we will choose what are the relevant columns to keep in the modelling and remove all irrelevant columns. Reducing the variables/columns can help in improving the performance and cost effectiveness. Variables that are related to the manufacturer or car model will be removed as these variables cannot be used in predictions. As mentioned above, we will create 2 datasets - data_fuel and data_hybrid, which contains fuel vehicles and

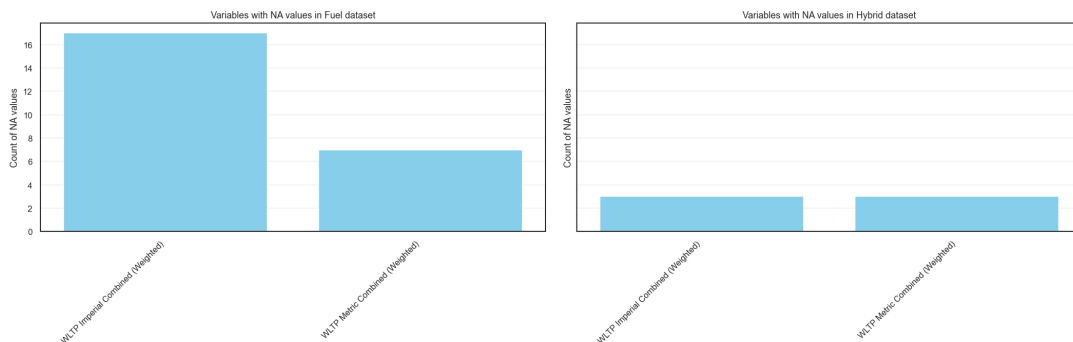
hybrid vehicles. Hence, the variables that are not relevant to fuel or hybrid will be removed too. For emissions, we will keep only Emissions NOx [mg/km] and WLTP CO2 as our target variables and drop rest of the emissions related columns. Lastly, date and standard category columns will be removed too as this information cannot be used for modeling.

Category	Keep	Remove
Vehicle	Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS)	Manufacturer, Model, Description, Transmission, Engine Power (Kw)
Fuel Consumption	WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined, WLTP Imperial Combined (Weighted), WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined, WLTP Metric Combined (Weighted)	Diesel VED Supplement, Testing Scheme
Electric Consumption		Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard Emission	Emissions NOx [mg/km], WLTP CO2	Euro Standard WLTP CO2 Weighted , THC Emissions [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date		Date of change

After removing the irrelevant variables, we have 19 variables remained in the datasets. Next, we will split our data into two groups as mentioned above which are the hybrid vehicles and fuel vehicles. We are getting **1686 hybrid** vehicles and **1250 fuel** vehicles in our datasets after splitting by the fuel type.

3.2 Data Cleaning

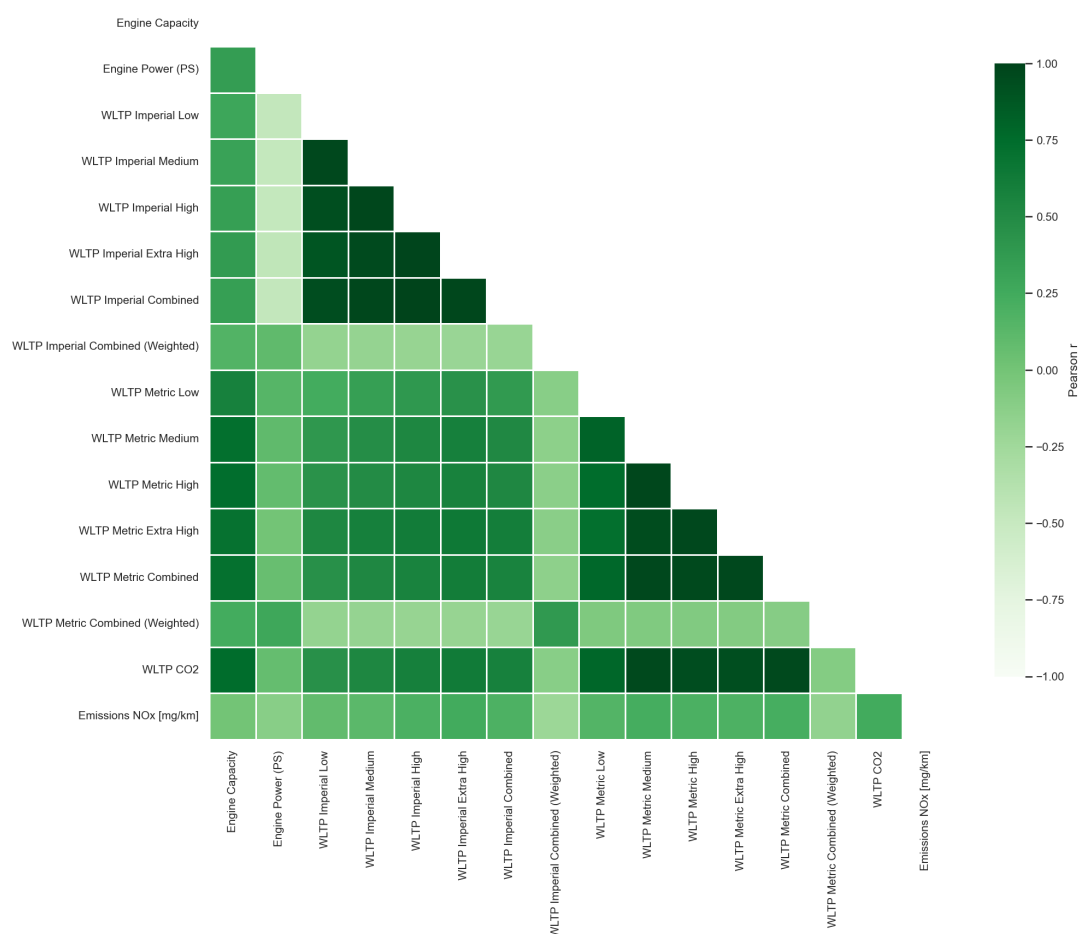
Before we start the modelling, cross checking on the data with “not available (NA)” values or empty, this will help to ensure the accuracy of our model. Based on the result, we found NA values in WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) columns in both fuel and hybrid datasets. After removing the NA rows, we have **1233 fuel** vehicles and **1683 hybrid** vehicles remained in the datasets.



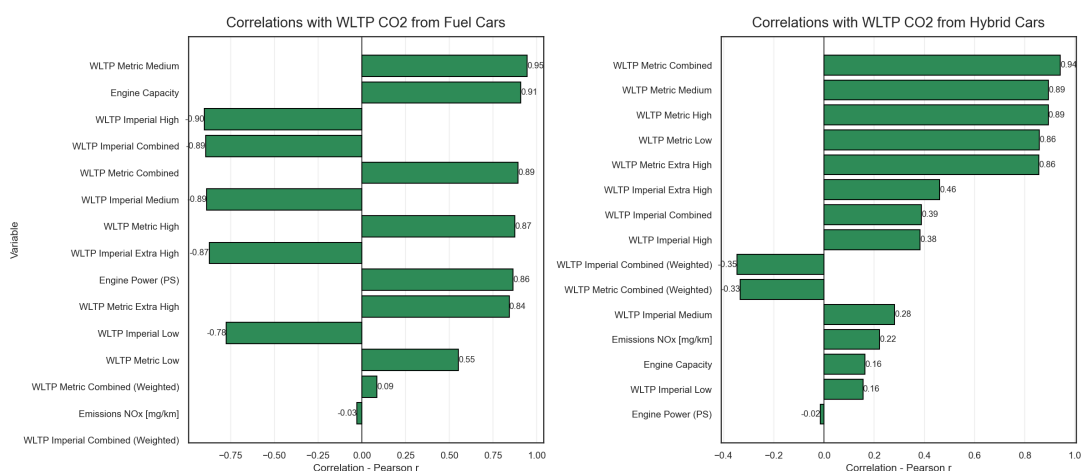
3.3 Correlation Analysis

Correlation analysis started with a heatmap to understand the correlation between each variables. From the heatmap, we can see the correlation with our target variables are low at Engine Power (Ps), WLTP Imperial Combined (Weighted), and WLTP Metric Combined (Weighted).

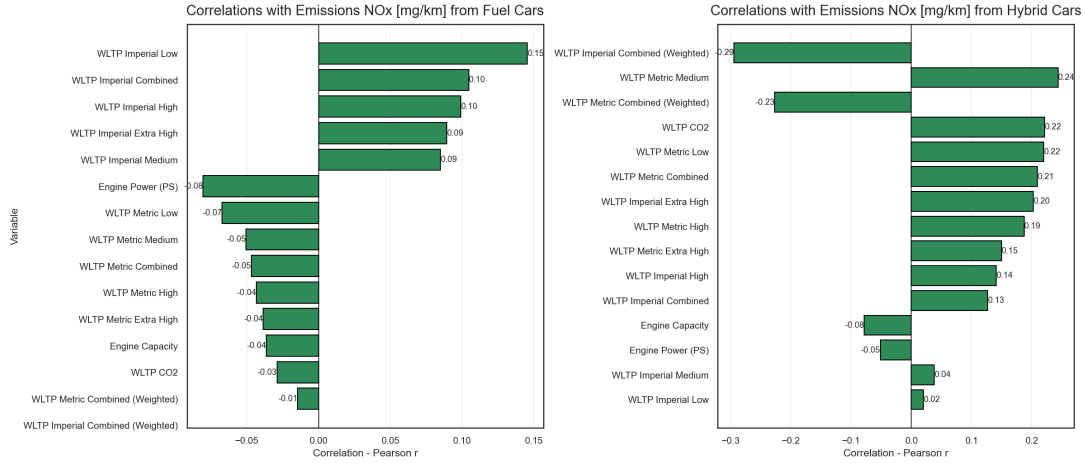
Correlation Heatmap (lower triangle)



When drilling down to the correlations with CO2, we can see that the WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) are having the lowest score in both fuel and hybrid vehicles. Whereas fuel is having high correlation with fuel vehicle but not the hybrid vehicle.



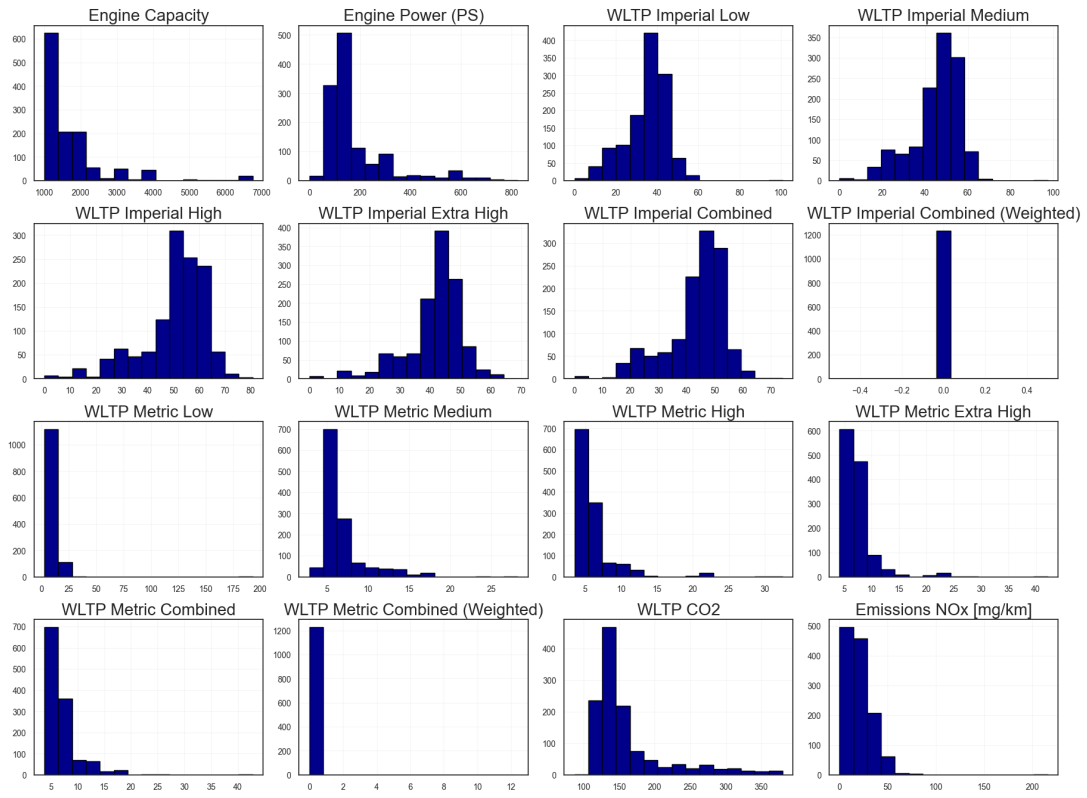
In NO_x emissions, WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) are still having lowest correlation in fuel vehicles but these variables are the top three correlated variables in hybrid vehicles. This could be due to the hybrid engine turns on intermittently, it often operates under non-steady, high-load conditions to recharge the battery or accelerate.



3.4 Data Distribution Analysis

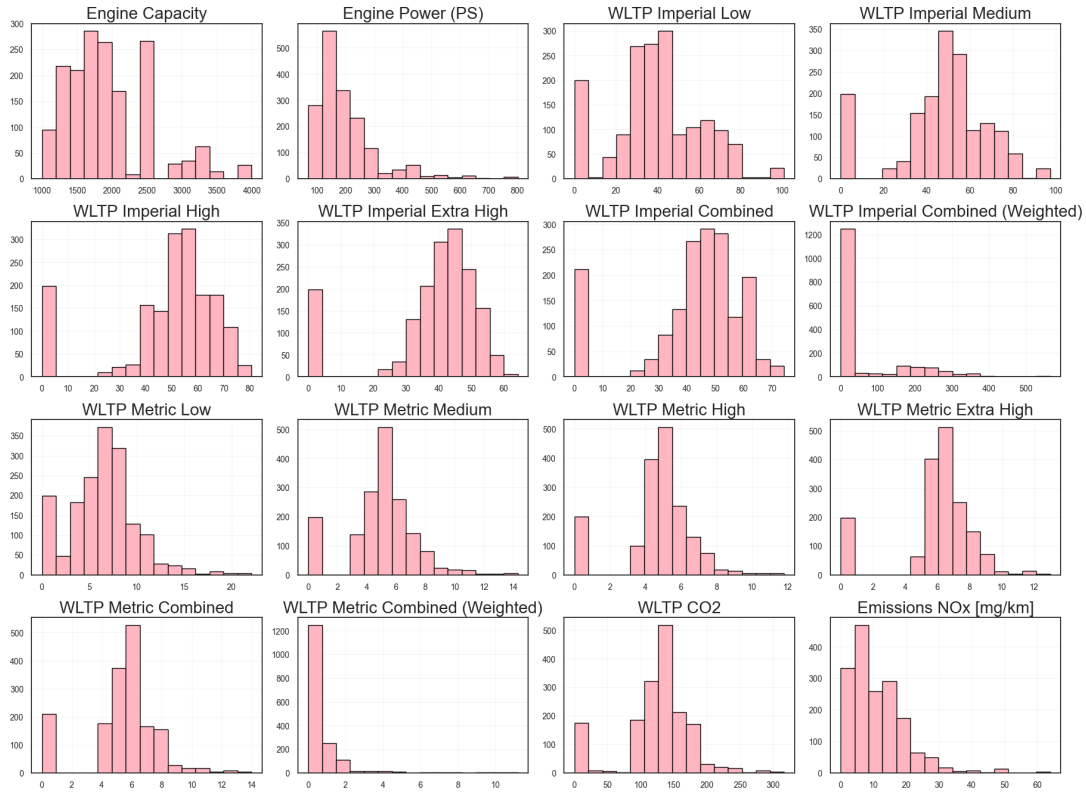
Distribution of data is important in identifying the normalisation or standardisation of each variables in our datasets. Below histogram matrix that showing the data distribution for fuel vehicles, we can see that WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) are not statistically normalised, as their disbutions are tightly clustered and do not follow a standardisation or z-score normalisation. Hence, we can safely remove WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) columns from *data_fuel* dataset

Histogram Matrix for Fuel Cars



Each variable in hybrid vehicles is statistically normalised, we do not see abnormal data distribution here. Hence, no changes to the variables for hybrid vehicles.

Histogram Matrix for Hyrbid Cars



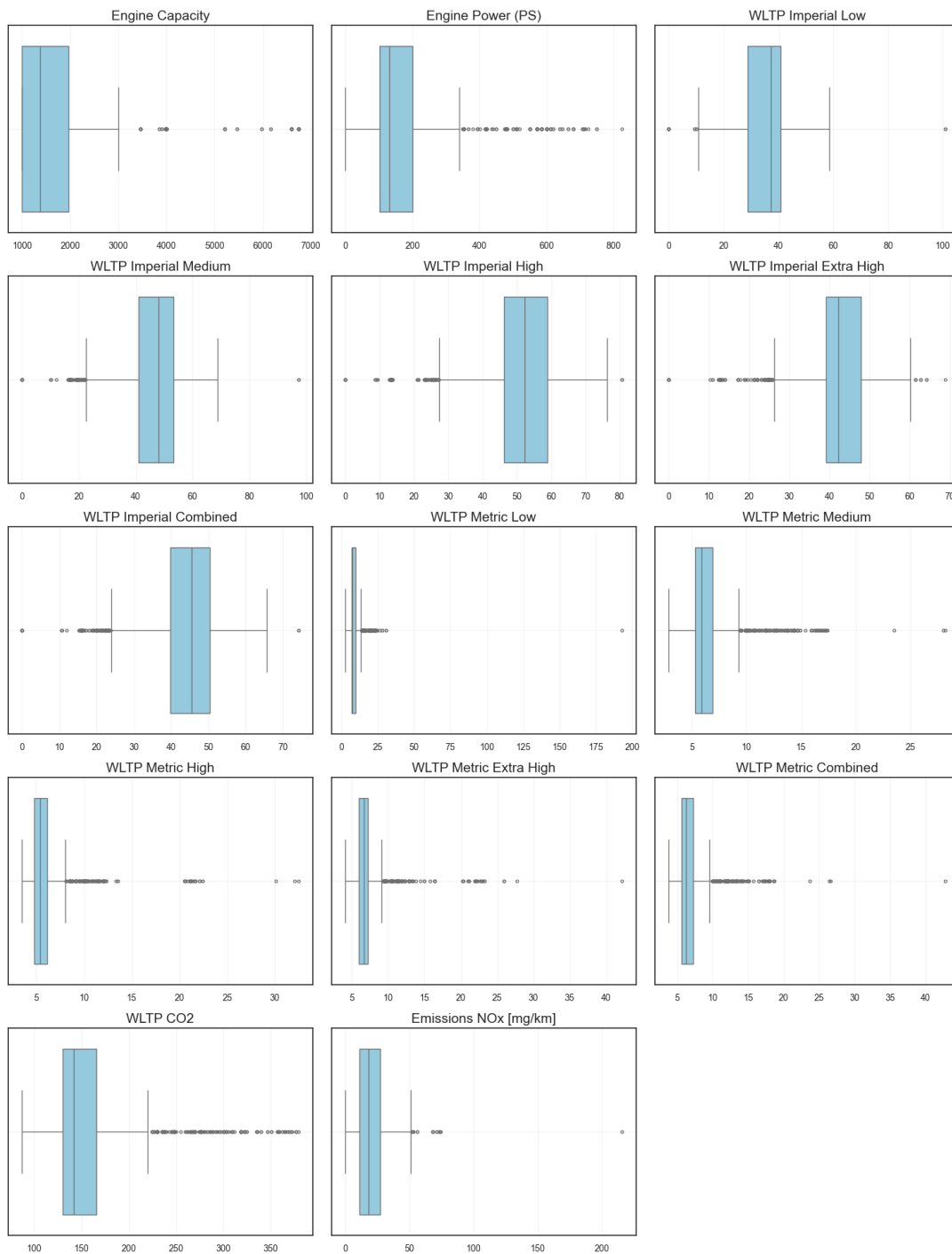
3.5 Outlier Check

To further improve our model and increase the accuracy, we have perform an analysis of the outliers for both datasets. In the fuel vehicles, we can clearly see the outliers are strongly skewed in most of the WLTP Metric variables and highly skewed in Engine Capacity and Power. This could be due to the high-performance models of vehicles. Since, the WLTP Imperial is having the same purpose with a different unit of measurements, we have decided to drop all the WLTP Metric variables which having higher outliers compare to MLTP Imperial. This table showing the updated model for *data_fuel*.

Category	Keep	Remove
Vehicle	Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS)	Manufacturer, Model, Description, Transmission, Engine Power (Kw)

Category	Keep	Remove
Fuel Consumption	WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined	Diesel VED Supplement, Testing Scheme, WLTP Imperial Combined (Weighted), WLTP Metric Combined (Weighted), WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined
Electric Consumption		Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard Emmission	Emissions NOx [mg/km], WLTP CO2	Euro Standard WLTP CO2 Weighted , THC Emissions [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date		Date of change

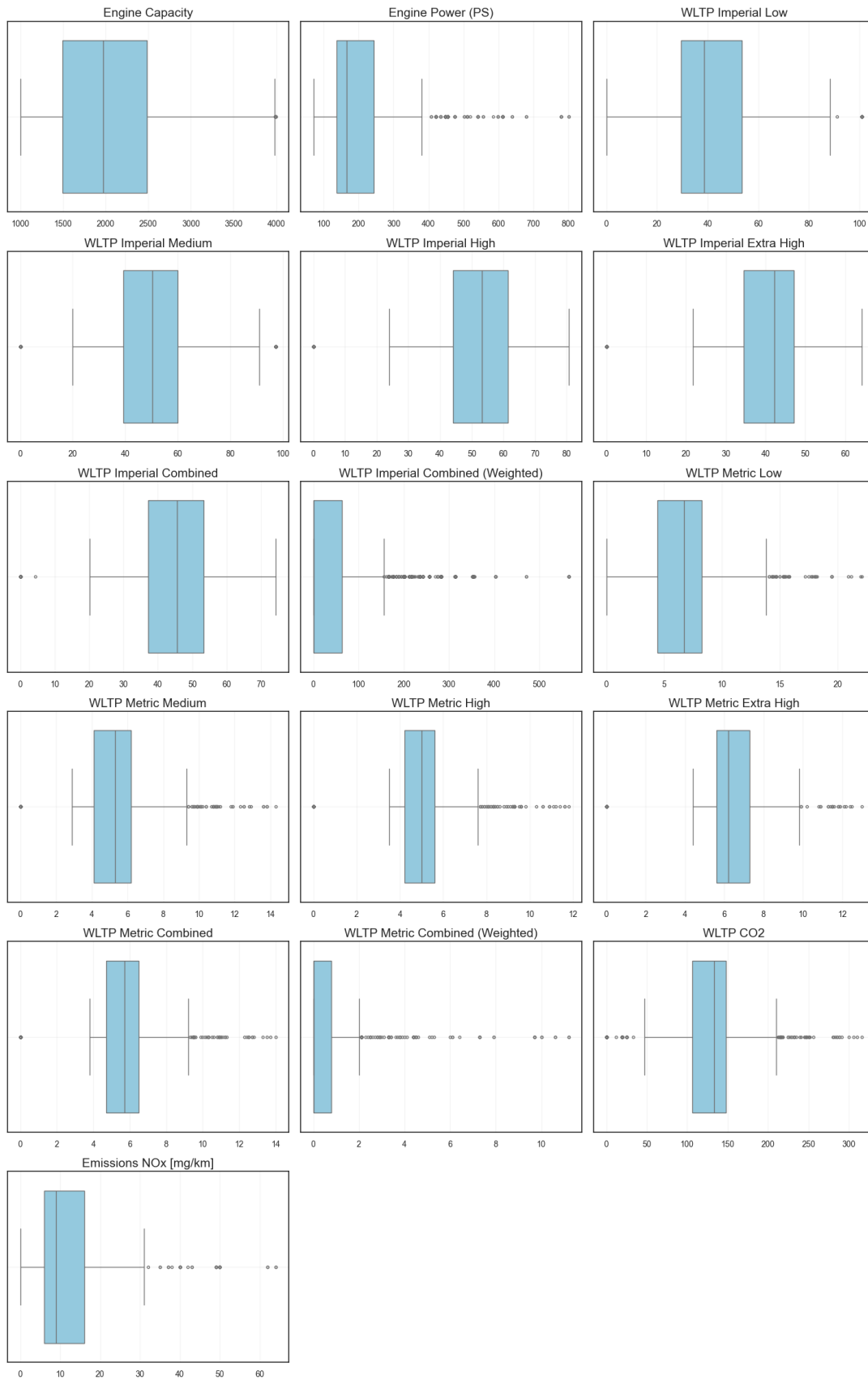
Outlier Visualisation (Boxplots)



Same for fuel vehicles, it seems to be a high outliers in WLTP Metric for hybrid vehicles. Therefore, these variables will be removed from hybrid vehicles dataset too. Here is the table showing the updated columns remain in *data_hybrid*.

Category	Keep	Remove
Vehicle	Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS)	Manufacturer, Model, Description, Transmission, Engine Power (Kw)
Fuel Consumption	WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined	Diesel VED Supplement, Testing Scheme, WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined, WLTP Metric Combined (Weighted), WLTP Imperial Combined (Weighted)
Electric Consumption		Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard		Euro Standard
Emmission	Emissions NOx [mg/km], WLTP CO2	WLTP CO2 Weighted , THC Emissions [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date		Date of change

Outlier Visualisation (Boxplots)



3.6 Cross Validation

Now, we have the final model with all the features and target variables to train and test and model. Before we start the model, a `data_split` function was created to split the `data_fuel` and `data_hybrid` into train, validate, and test datasets. The datasets were split into **80% for training, 10% for validations and 10% for testing**.

After splitting the data, we have created a function called `train_lasso_with_cv` to use **Lasso** model for feature selection with a **10-fold** cross validations. Based on the results from Lasso model, we have designed to refine the Lasso model again with the best L1 regularisation strength (*alphas*). With the results from Lasso model, we can understand the errors using **mean squared error (MSE)** and variances by calculating the **r2_score** as well as listing the coefficient of each features.

We have done the training for both datasets `data_fuel` and `_data_hybrid` with the two target variables *WLTP CO2* and *Emissions NOx [mg/km]* that we have defined above. Below table is the results from the *lasso* model training with *10-fold* cross validation.

Metric	Fuel / CO2 (12 fea- tures)	Fuel / CO2 (3 fea- tures)	Fuel / NOx (12 fea- tures)	Fuel / NOx (3 fea- tures)	Hybrid / CO2 (12 features)	Hybrid / CO2 (3 fea- tures)	Hybrid / NOx (12 features)	Hybrid / NOx (3 fea- tures)
Best alpha	0.054	0.022	0.022	0.002	0.103	0.022	0.004	0.055
Train RMSE	197.082	2071.979	213.583	238.598	1359.811	2071.979	45.390	74.931
Validate RMSE	270.483	1912.145	1272.888	171.254	1044.115	1912.145	71.701	98.815
Validate R ²	0.926	0.055	-6.459	-0.003	0.484	0.055	0.279	0.007
Test RMSE	247.201	2080.948	1172.173	149.236	1525.036	2080.948	24.205	45.544
Test R ²	0.890	0.251	6.831	0.002	0.451	0.251	0.463	-0.009

Based on the *alphas* returned from first round of training, we have run the *lasso* model again with following results.

Metric (re- fined)	Fuel / CO2 (12 fea- tures)	Fuel / CO2 (3 fea- tures)	Fuel / NOx (12 fea- tures)	Fuel / NOx (3 fea- tures)	Hybrid / CO2 (12 features)	Hybrid / CO2 (3 fea- tures)	Hybrid / NOx (12 features)	Hybrid / NOx (3 fea- tures)
Validate RMSE	261.047	1912.280	1276.104	171.228	1043.926	1912.280	71.691	98.813
Validate R ²	0.928	0.055	-6.478	-0.003	0.484	0.055	0.279	0.007
Test RMSE	236.887	2081.085	1175.241	149.221	1525.295	2081.085	24.225	45.544
Test R ²	0.894	0.251	-6.852	0.003	0.451	0.251	0.462	-0.009

From the *lasso* training, we found that the model with all the 12 features are working fine in both datasets for fuel and hybrid vehicle when we examine the correlation with CO₂ and NO_x. The errors have increased drastically after reduced to 3 features only. Eventhough the model for Fuel/NO_x was strongly negative in validation R², we are keeping the 12 features to standardised the features selection. We will consider an algorithm improvements in the future.

Next, we have done a diagnostic using **residual/prediction scatter plot** to check the scalar metrics on the 12 features.

3.6.1 Fuel / WLTP CO₂

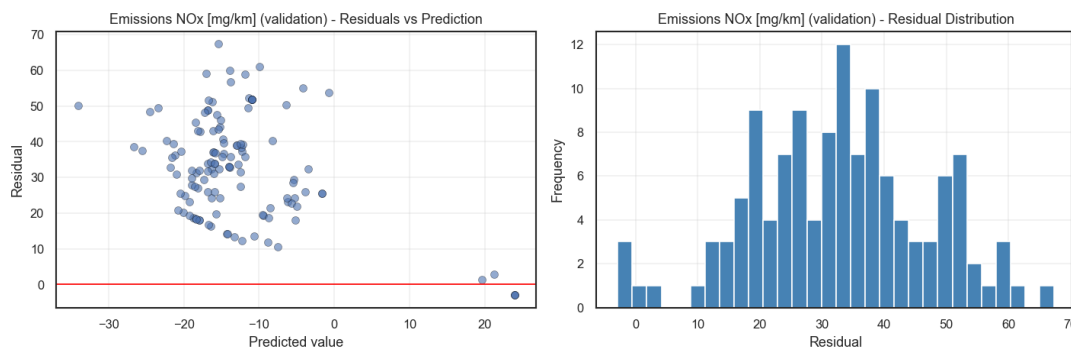
Residual scatter is tight, centered on zero, and roughly uniform across fitted values. The histogram is symmetric with light tails that matches the small residual mean/std (~0, 14) and strong validation R² (~0.93), so the model is well calibrated with no obvious bias or variances.

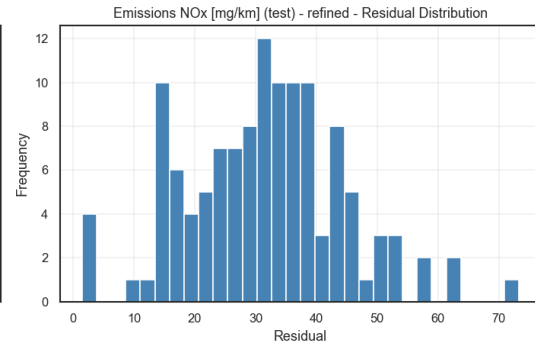
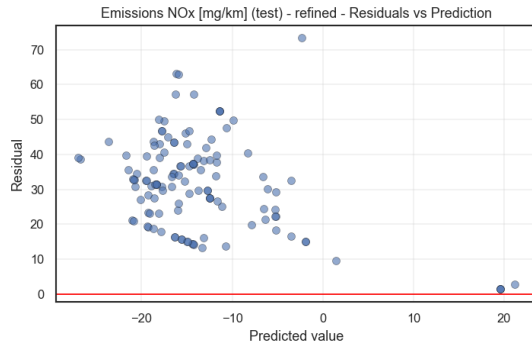
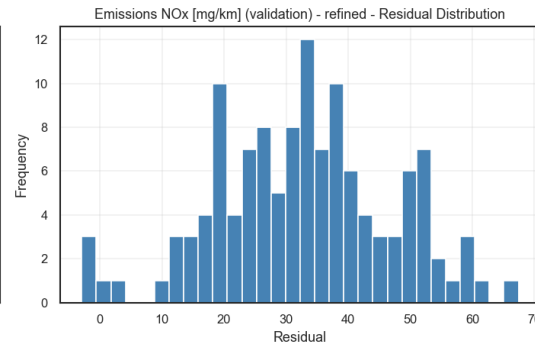
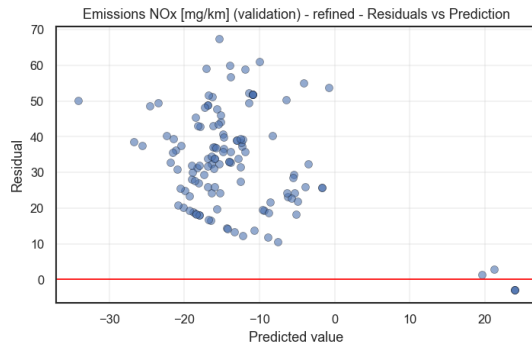
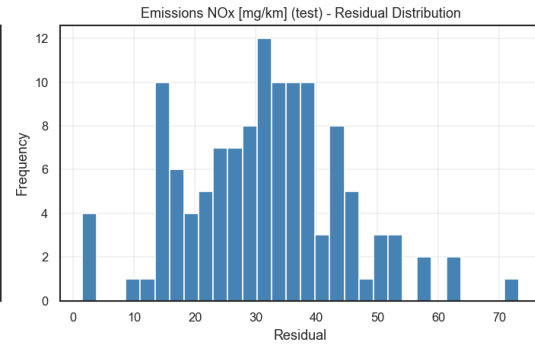
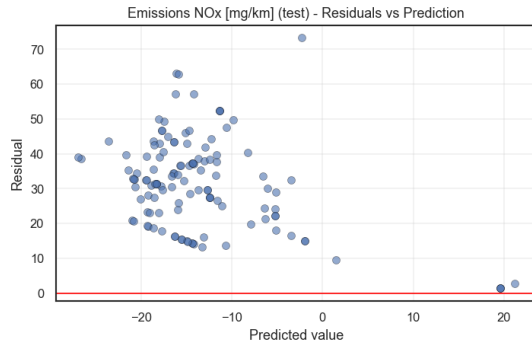
Best	Train RMSE	Val RMSE	Val R ²	Test RMSE	Test R ²
0.054	14.039	16.446	0.926	15.723	0.890
2.976	13.994	24.107	0.840	24.144	0.740

Best	Train RMSE	Val RMSE	Val R ²	Test RMSE	Test R ²
0.143	14.786	14.781	0.940	13.299	0.921
2.976	14.787	14.790	0.940	13.343	0.921

3.6.2 Fuel / NO_x

Residuals skew sharply positive (validation residuals mostly above zero, maxima > +180). The model seems to be under-predicts high NO_x readings, which is why validation/test R² plunge well below zero. The histogram's heavy right tail confirms the bias and we might need to add more features or change a different model for fuel vehicles and NO_x emission prediction.



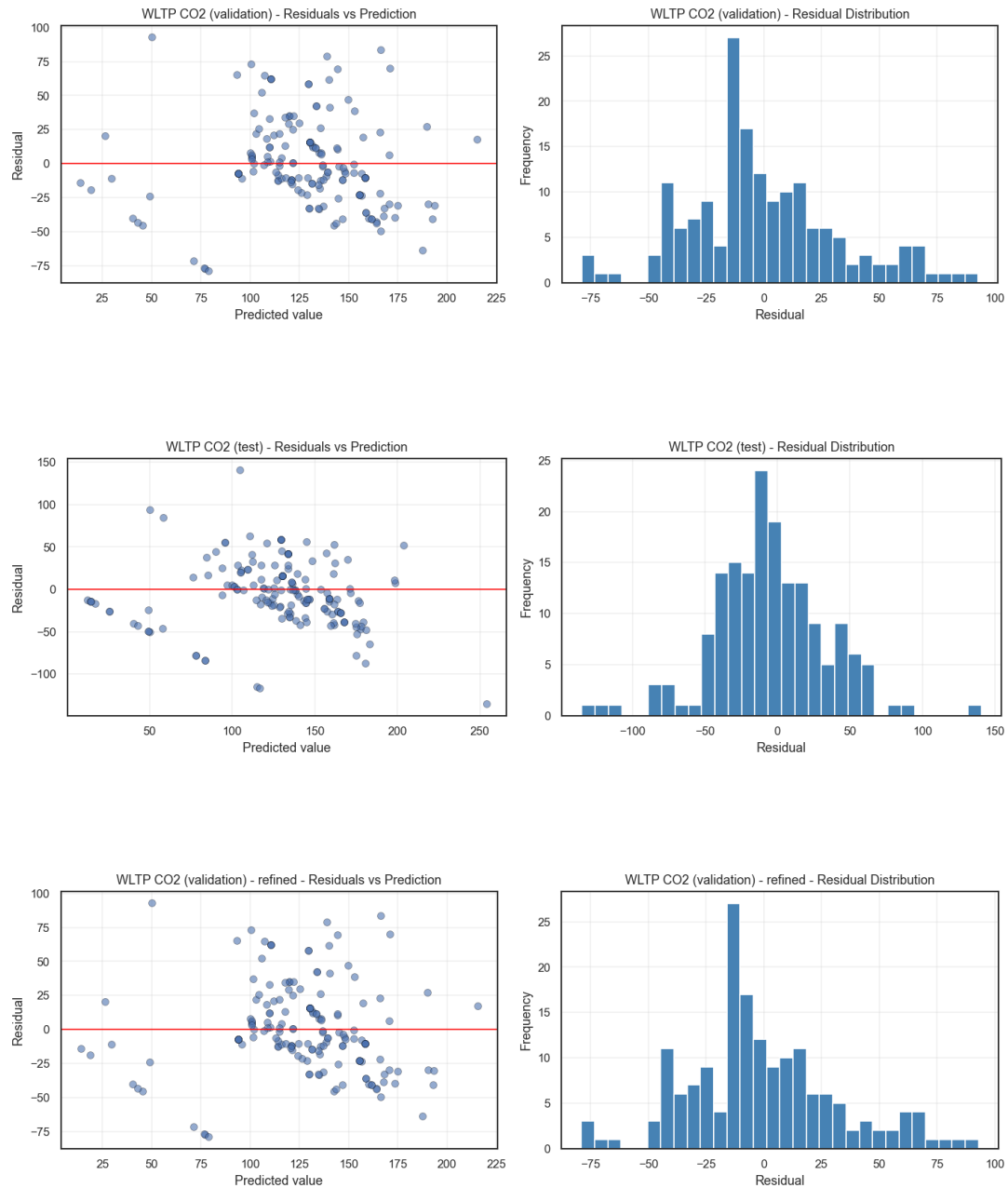


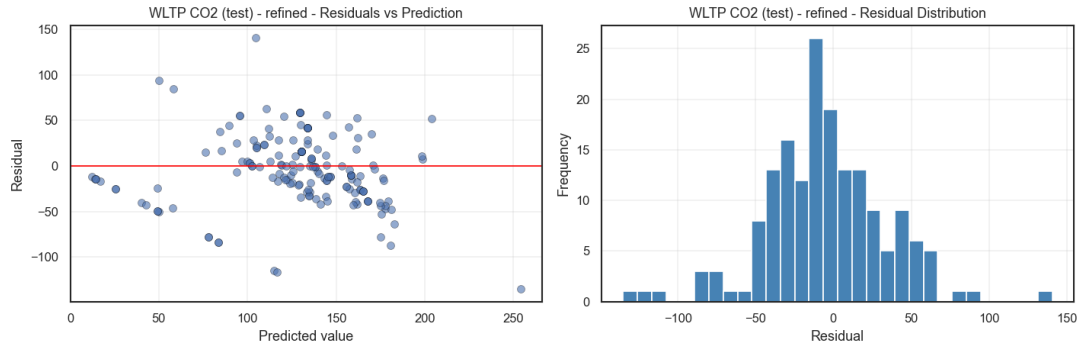
Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
0.023	14.614	35.678	-6.460	34.237	-6.832
2.976	14.613	36.945	-6.999	35.501	-7.421

Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
0.002	15.447	13.086	-0.004	12.216	0.003
233.572	15.479	12.929	0.020	12.133	0.016

3.6.3 Hybrid / CO2

Residual cloud is symmetric but very wide (± 140), showing high variance even though it remains centered. That matches RMSE $\sim 32\text{--}39$ and explains only $\sim 45\text{--}48\%$ of variance, we might consider adding more features or a nonlinear terms to lower down the variance.



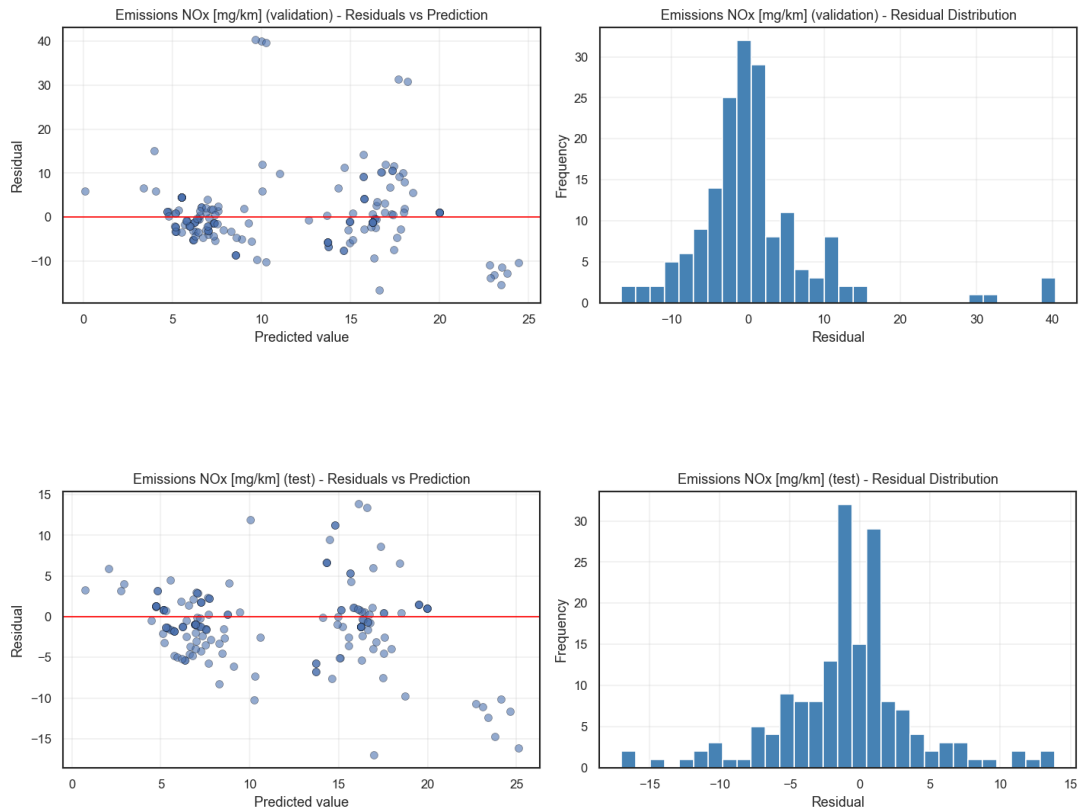


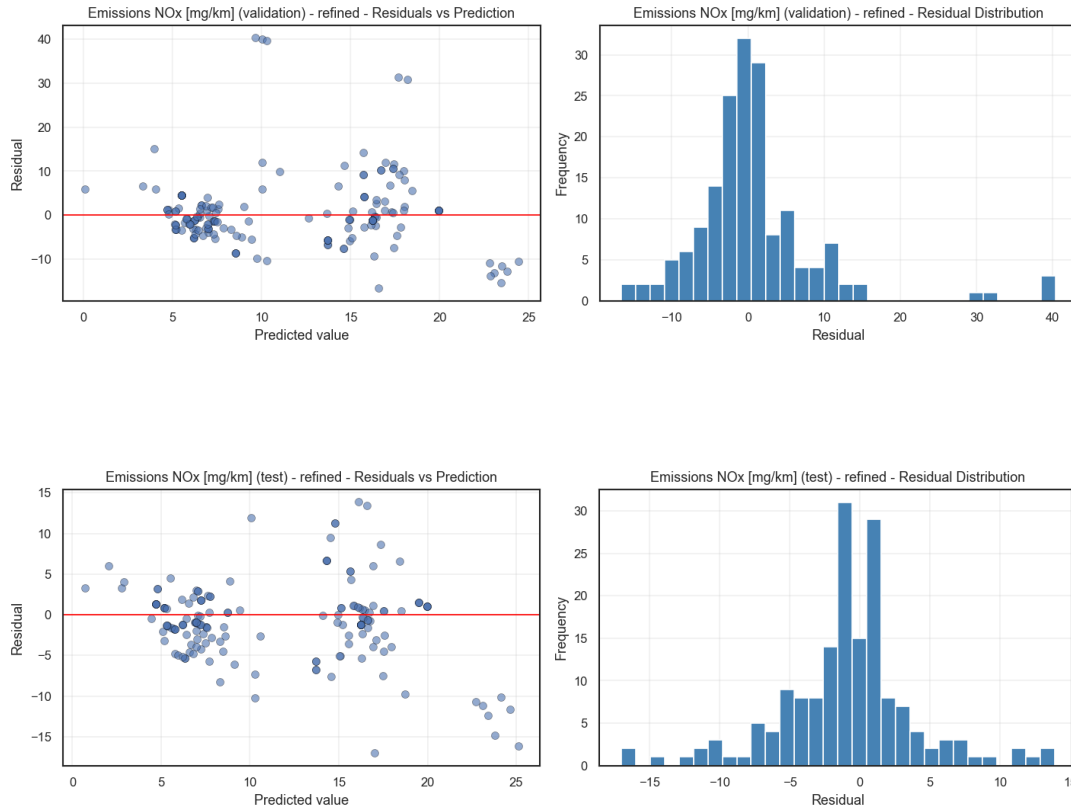
Best	Train RMSE	Val RMSE	Val R ²	Test RMSE	Test R ²
0.103	36.876	32.313	0.484	39.052	0.451
6.158	36.881	32.256	0.486	39.147	0.448

Best	Train RMSE	Val RMSE	Val R ²	Test RMSE	Test R ²
0.022	45.519	43.728	0.055	45.617	0.251
6.158	45.520	43.691	0.057	45.595	0.252

3.6.4 Hybrid / NOx

Residuals are compact, centered, and evenly spread, giving a clean plot with no obvious bias. Validation/test R² around 0.28–0.46 confirm a solid fit.





Best	Train RMSE	Val RMSE	Val R ²	Test RMSE	Test R ²
0.005	6.737	8.468	0.279	4.920	0.463
26.367	6.752	8.481	0.277	4.805	0.488

Best	Train RMSE	Val RMSE	Val R ²	Test RMSE	Test R ²
0.056	8.656	9.941	0.007	6.749	-0.010
1000.000	8.658	9.968	0.002	6.746	-0.009

Overall, the residual diagnostics show the CO₂ models behave reasonably especially in the fuel dataset, the hybrid CO₂ models suffer from high unexplained variance, and the fuel NO_x model with all features is badly biased either transform features or try an alternative modeling approach to capture the heavy right tail.

4 Neural Network - Regression Tasks

Variables Selection The following variables should be used to develop neural networks model.

As this is regression tasks, hidden layers will be applied with ReLU - max(0,x) as activation method to keep gradients strong and captures nonlinearity. The output layer will apply linear

approach without activation to allow continuous outputs from the gas emissions, with no upper and lower limit

Parameters to train all the model: * Step size: 0.004. To sufficient enough to learn. Small values like 0.00005 can be too high depending on your data scale * Iteration: 2,001. To * Number of nodes in each layer: 10-25 nodes.

4.1 Neural Network Regression (2 Layers)

Training RMSE: 0.00782664077441073

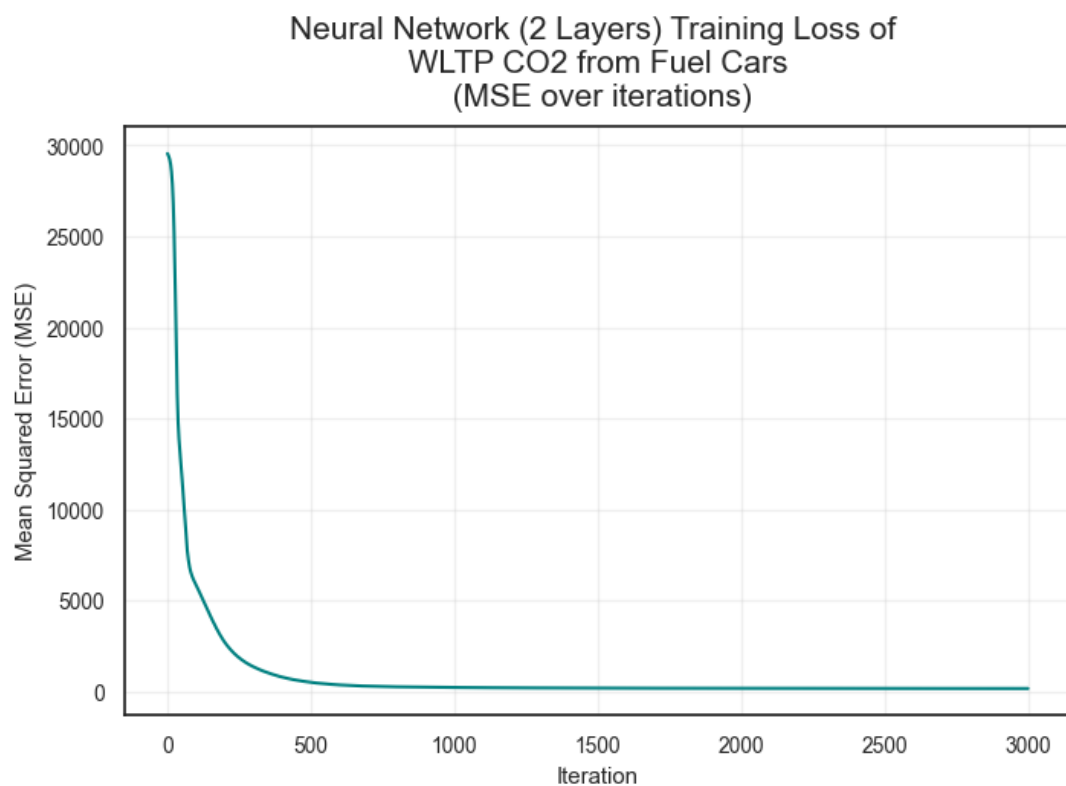
Coefficient of determination (R^2) on training data: 1.0

Test RMSE: 69.79849360200608

Coefficient of determination (R^2) on test data: -1.1719

Validation RMSE: 72.88376485845356

Coefficient of determination (R^2) on validation data: -0.4593



Training RMSE: 0.0010511071694817727

Coefficient of determination (R^2) on training data: 1.0

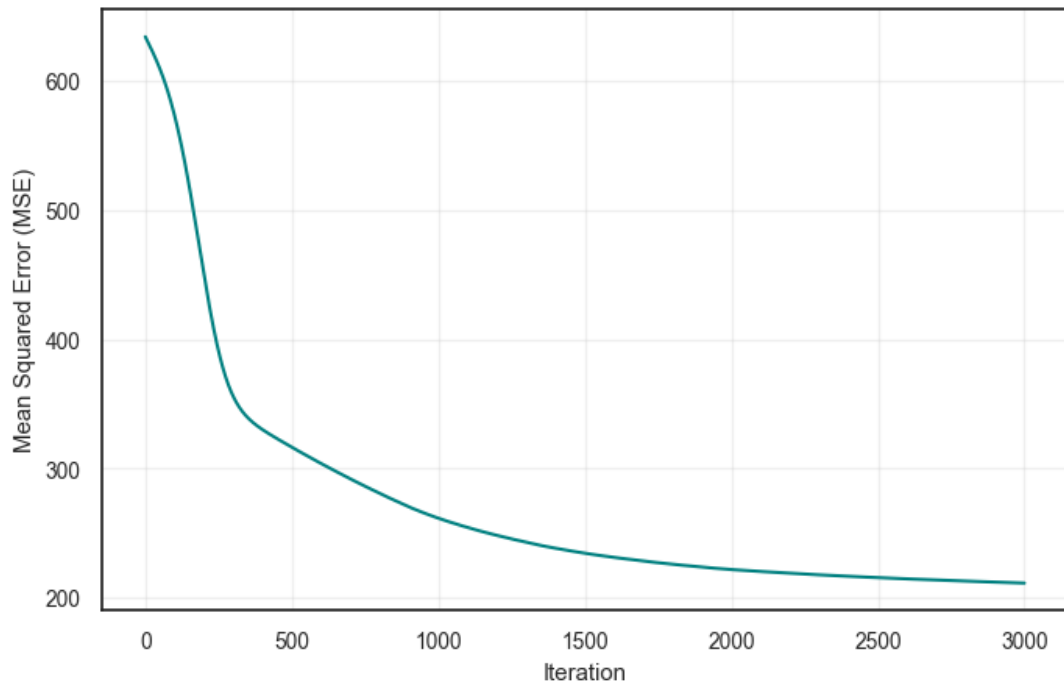
Test RMSE: 14.862229155978921

Coefficient of determination (R^2) on test data: -0.4758

Validation RMSE: 16.46651263704182

Coefficient of determination (R^2) on validation data: -0.5891

Neural Network (2 Layers) Training Loss of
Emissions NOx [mg/km] from Fuel Cars
(MSE over iterations)



Training RMSE: 0.0047847548570770625

Coefficient of determination (R^2) on training data: 1.0

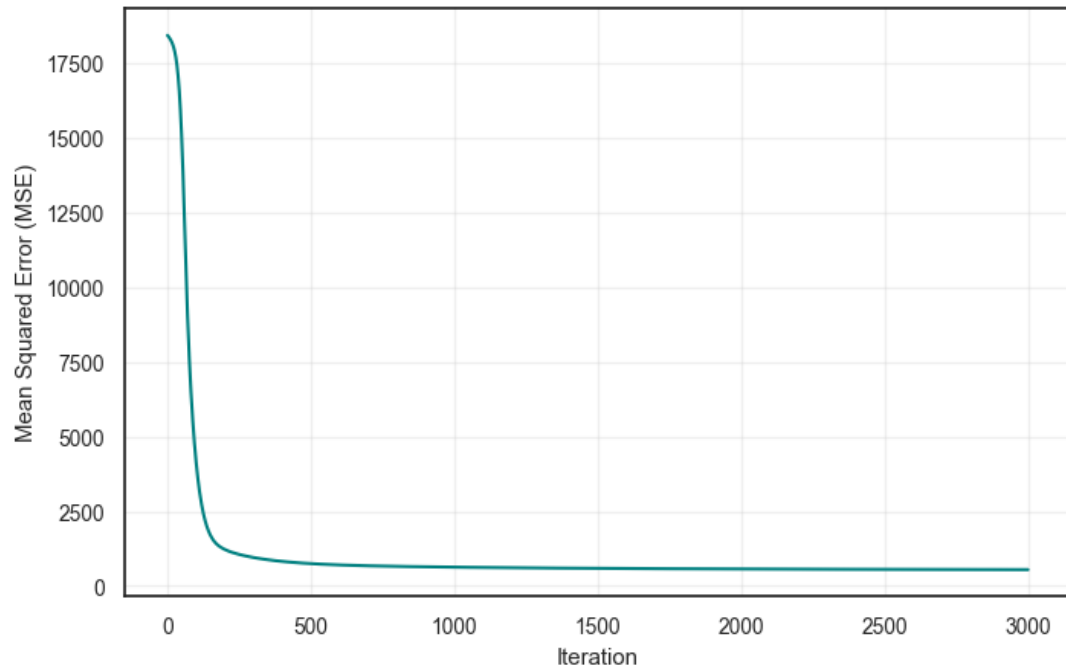
Test RMSE: 27.859055280851404

Coefficient of determination (R^2) on test data: 0.7206

Validation RMSE: 18.148562709938737

Coefficient of determination (R^2) on validation data: 0.8373

Neural Network (2 Layers) Training Loss of
WLTP CO2 from Hybrid Cars
(MSE over iterations)



Training RMSE: 0.0005385192979172459

Coefficient of determination (R^2) on training data: 1.0

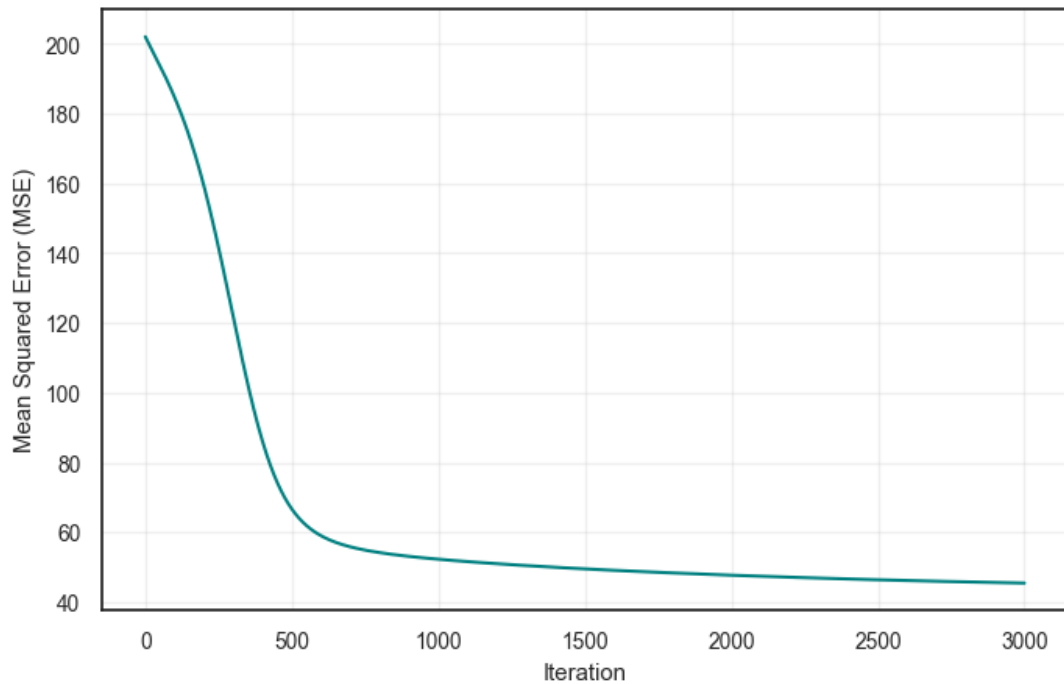
Test RMSE: 4.646769287505099

Coefficient of determination (R^2) on test data: 0.5212

Validation RMSE: 8.297508038655964

Coefficient of determination (R^2) on validation data: 0.3081

Neural Network (2 Layers) Training Loss of
Emissions NOx [mg/km] from Hybrid Cars
(MSE over iterations)



4.2 Neural Network Regression - 3 Layers

Training RMSE: 5.726885806751038

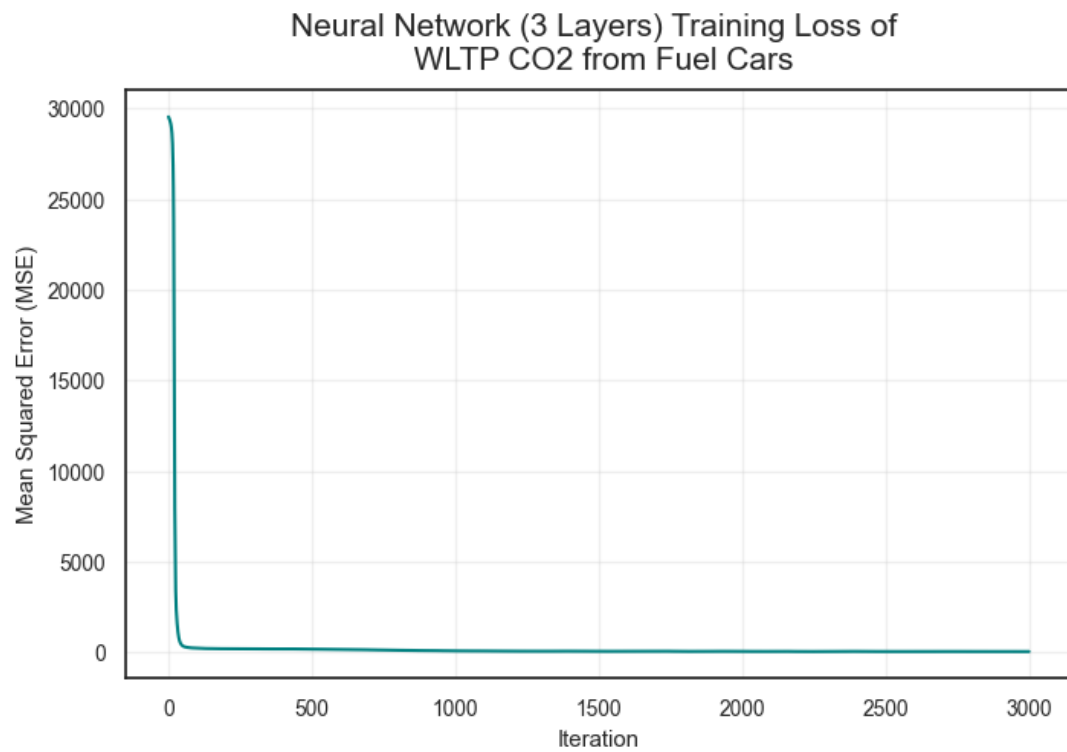
Coefficient of determination (R^2) on training data: 0.9892

Test RMSE: 61.25398062710987

Coefficient of determination (R^2) on test data: -0.6727

Validation RMSE: 61.62508793707413

Coefficient of determination (R^2) on validation data: -0.0433



Training RMSE: 14.015438850518084

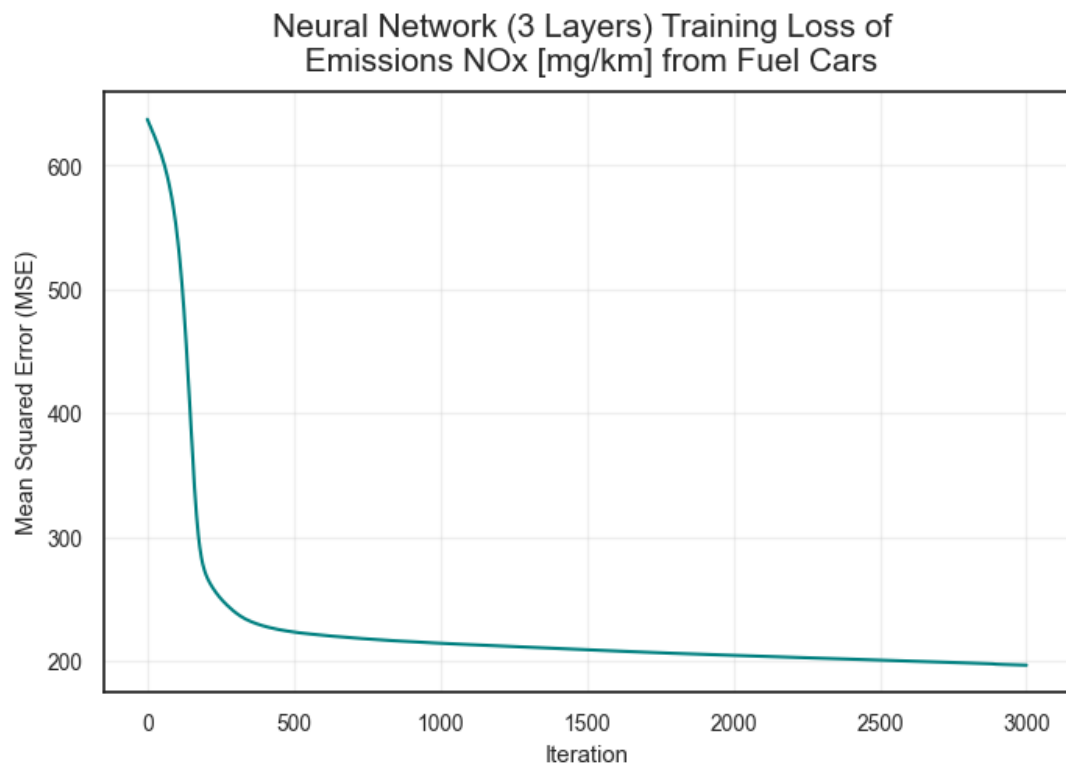
Coefficient of determination (R^2) on training data: 0.1991

Test RMSE: 16.430205934485976

Coefficient of determination (R^2) on test data: -0.8036

Validation RMSE: 18.211237381907758

Coefficient of determination (R^2) on validation data: -0.9437



Decide which variables to be removed for fuel cars (comment here)

Training RMSE: 16.844755984845527

Coefficient of determination (R^2) on training data: 0.9032

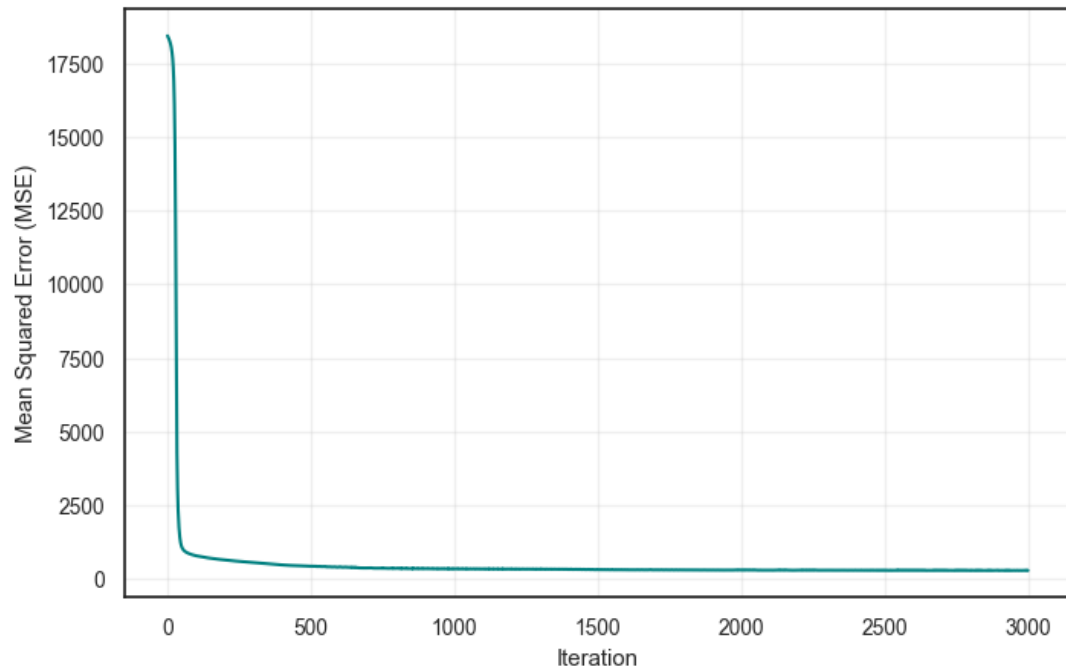
Test RMSE: 16.96588022436222

Coefficient of determination (R^2) on test data: 0.8964

Validation RMSE: 11.594324025650721

Coefficient of determination (R^2) on validation data: 0.9336

Neural Network (3 Layers) Training Loss of
WLTP CO2 from Hybrid Cars



Training RMSE: 6.504473746098478

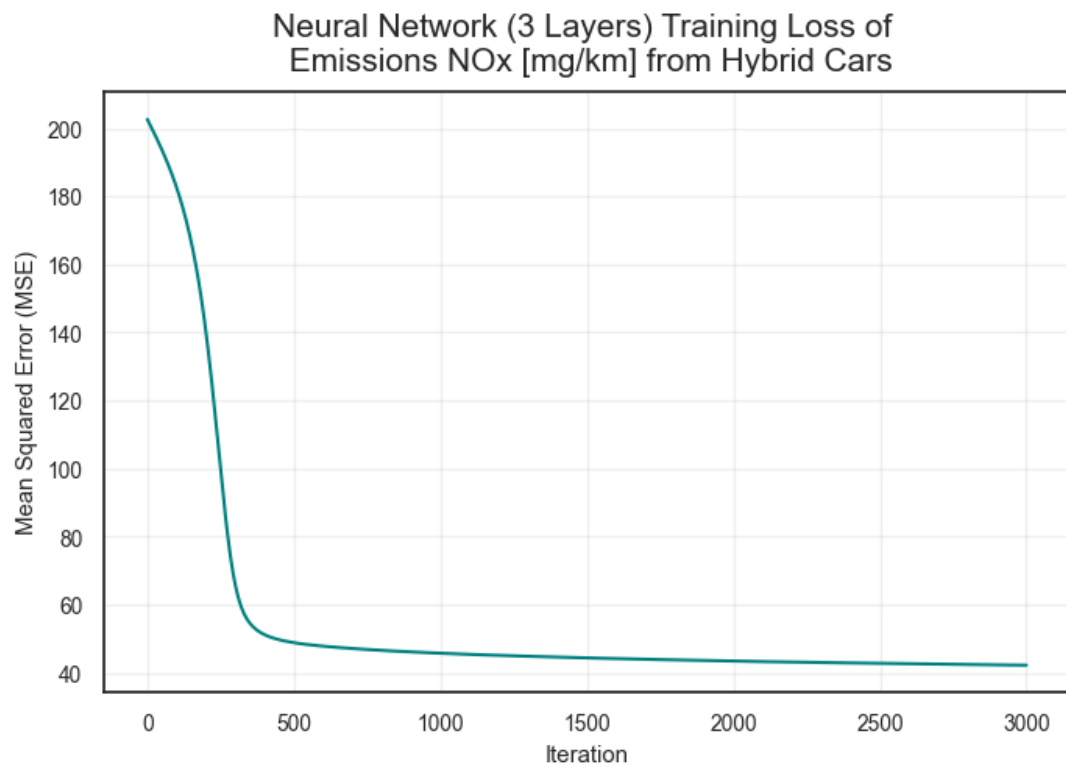
Coefficient of determination (R^2) on training data: 0.4379

Test RMSE: 4.777773727152571

Coefficient of determination (R^2) on test data: 0.4938

Validation RMSE: 8.237238874814592

Coefficient of determination (R^2) on validation data: 0.3182



4.3 Neural Network Regression - 4 Layers

Training RMSE: 8.069100323986483

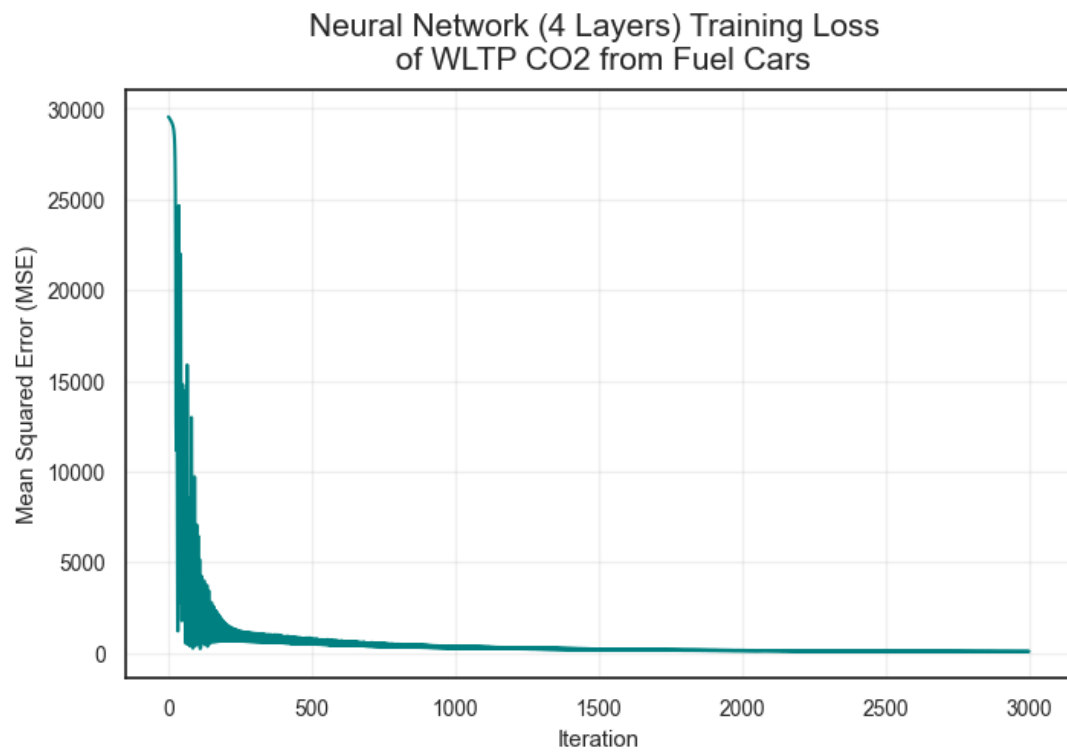
Coefficient of determination (R^2) on training data: 0.9787

Test RMSE: 23.89181215635021

Coefficient of determination (R^2) on test data: 0.7455

Validation RMSE: 24.803493834032274

Coefficient of determination (R^2) on validation data: 0.831



Training RMSE: 12.853882671544087

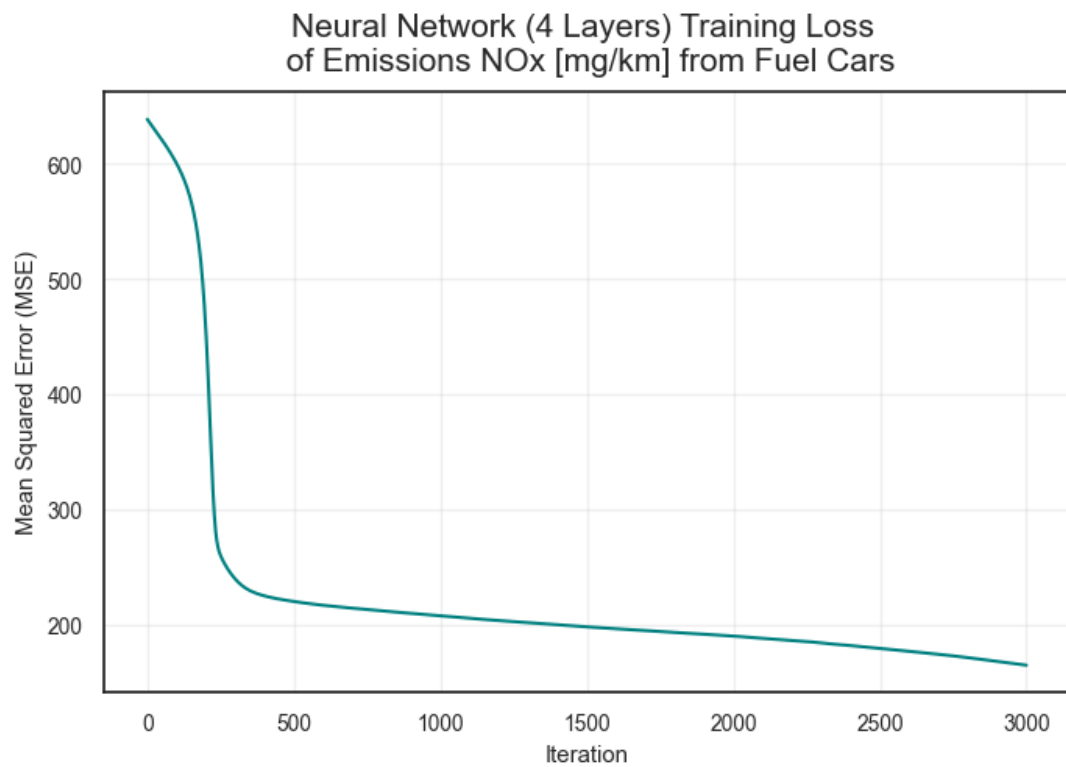
Coefficient of determination (R^2) on training data: 0.3263

Test RMSE: 18.305318574045142

Coefficient of determination (R^2) on test data: -1.2388

Validation RMSE: 19.945129432398396

Coefficient of determination (R^2) on validation data: -1.3314



Training RMSE: 16.126670497599502

Coefficient of determination (R^2) on training data: 0.9113

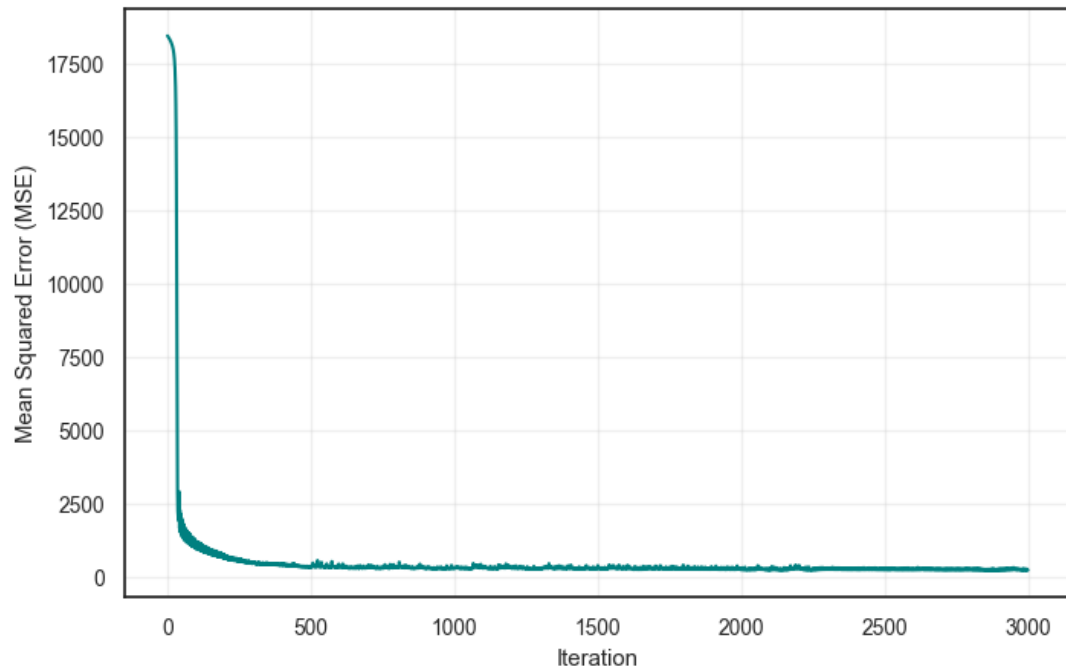
Test RMSE: 15.388179581929549

Coefficient of determination (R^2) on test data: 0.9148

Validation RMSE: 11.27180118070615

Coefficient of determination (R^2) on validation data: 0.9372

Neural Network (4 Layers) Training Loss
of WLTP CO2 from Hybrid Cars



Training RMSE: 6.34412972600985

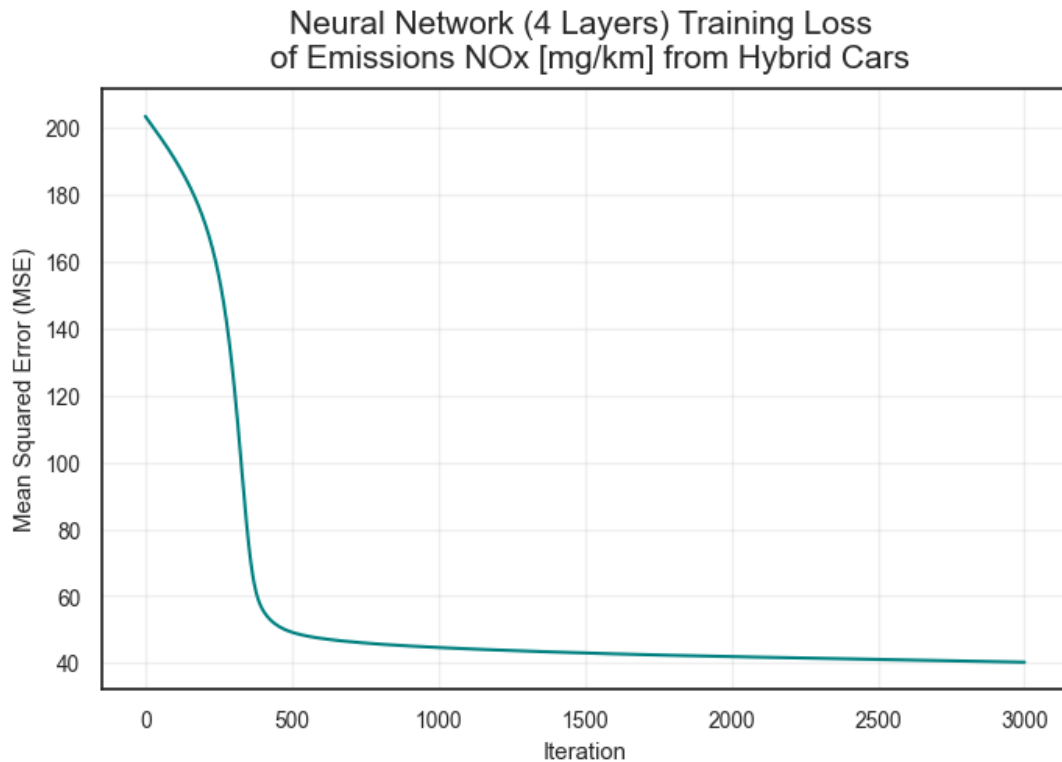
Coefficient of determination (R^2) on training data: 0.4653

Test RMSE: 4.655104021915579

Coefficient of determination (R^2) on test data: 0.5195

Validation RMSE: 8.134898811912487

Coefficient of determination (R^2) on validation data: 0.335



5 Support Vector Machine - Regression

6 References

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